

Yibo Yin

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EDUCATION

École polytechnique fédérale de Lausanne (EPFL)

Lausanne, Switzerland

M.Sc. in Digital Humanities

09/2025 - Present

- Research Focus: Computer Graphics and Computer Vision
- GPA: 5.56 /6.00

Wuhan University

Wuhan, China

B.Eng. in Computer Science and Technology

09/2021 - 06/2025

- GPA: 3.93 /4.00 (top 4%)

PUBLICATIONS

MATStruct: High-Quality Medial Mesh Computation via Structure-aware Variational Optimization

Ningna Wang, Rui Xu, **Yibo Yin**, Zichun Zhong, Taku Komura, Wenping Wang, Xiaohu Guo
ACM SIGGRAPH Asia 2025 (Conference Track)

RESEARCH EXPERIENCE

Image and Visual Representation Lab, EPFL

06/2026 - Present

Research Assistant | PI: Prof. Sabine Süsstrunk [✉](#); Supervisor: Zhuoqian Yang, Doctoral Assistant [✉](#)

- Investigating weight space generation for implicit neural representations, with the goal of learning modality-agnostic priors over neural fields across multiple data modalities.

Realistic Graphics Lab, EPFL

03/2026 - 06/2026

Research Assistant | Advisor: Prof. Wenzel Jakob [✉](#)

- Developed a differentiable Gaussian Process Implicit Surface (GPIS) sampling module in Dr.Jit to support inverse-rendering with implicit geometry.
- Developed a hybrid surface-volume rendering approach for Many-Worlds inverse rendering.

Computer Graphics Lab, The University of Texas at Dallas

04/2024 - 11/2025

Research Assistant (Remote) | Advisors: Prof. Xiaohu Guo [✉](#) and Dr. Ningna Wang [✉](#)

- Adapted Loop subdivision surface evaluation and sharp feature fitting algorithms to structured medial mesh.
- Designed a simplification algorithm for structured medial mesh to preserve geometric structure under subdivision.
- Proposed a subdivision surface fitting method for structured medial mesh, targeting sharp-feature preservation during fitting.

Graphics and Vision Lab, Wuhan University

05/2023 - 01/2024

Research Assistant | Advisor: Prof. Chunxia Xiao [✉](#)

- Built an RGB-ToF/depth data acquisition pipeline and collected 10,000+ paired image-sensor samples.
- Proposed a novel monocular depth estimation method using RGB images and auxiliary ToF/depth signals.

SELECTED PROJECTS

Physically Based Ray Tracer

2026

CS-440 Advanced Computer Graphics Rendering Competition, EPFL

- Extended the Nori framework with heterogeneous volumetric path tracing, delta and ratio tracking, Henyey-Greenstein phase functions, and MIS over emitter, phase, and BSDF sampling.
- Implemented HDR environment map lighting, thin-lens depth of field, motion blur, principled BSDF, and GGX rough dielectric materials.

AWARDS

- **Outstanding Student** (10% school-wide), Wuhan University 2022, 2023, 2024
- **Second Class Scholarship** (10% school-wide), Wuhan University 2022
- **Third Class Scholarship** (15% school-wide), Wuhan University 2023, 2024
- **Lei Jun (Xiaomi) Computer Innovation and Development Fund**, Wuhan University 2024

SKILLS

- **Languages:** Mandarin (Native), English (TOEFL iBT 102)
- **Programming Languages:** C++, C, Python, C#, GLSL, SQL, Verilog HDL, Java, $\text{\LaTeX}$
- **Libraries / Frameworks / Tools:** PyTorch, Dr.Jit, Mitsuba, PBRT-v3, Blender, Eigen, CGAL, libigl, Git, CMake